



Ninga Grocery IA Project

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Abstract

This project is about studying the information structure (IA) of the Ninja mobile app, which is made for grocery deliver. The main goal is to check and improve the way the app is organized so that users can have a smooth and easy experience.

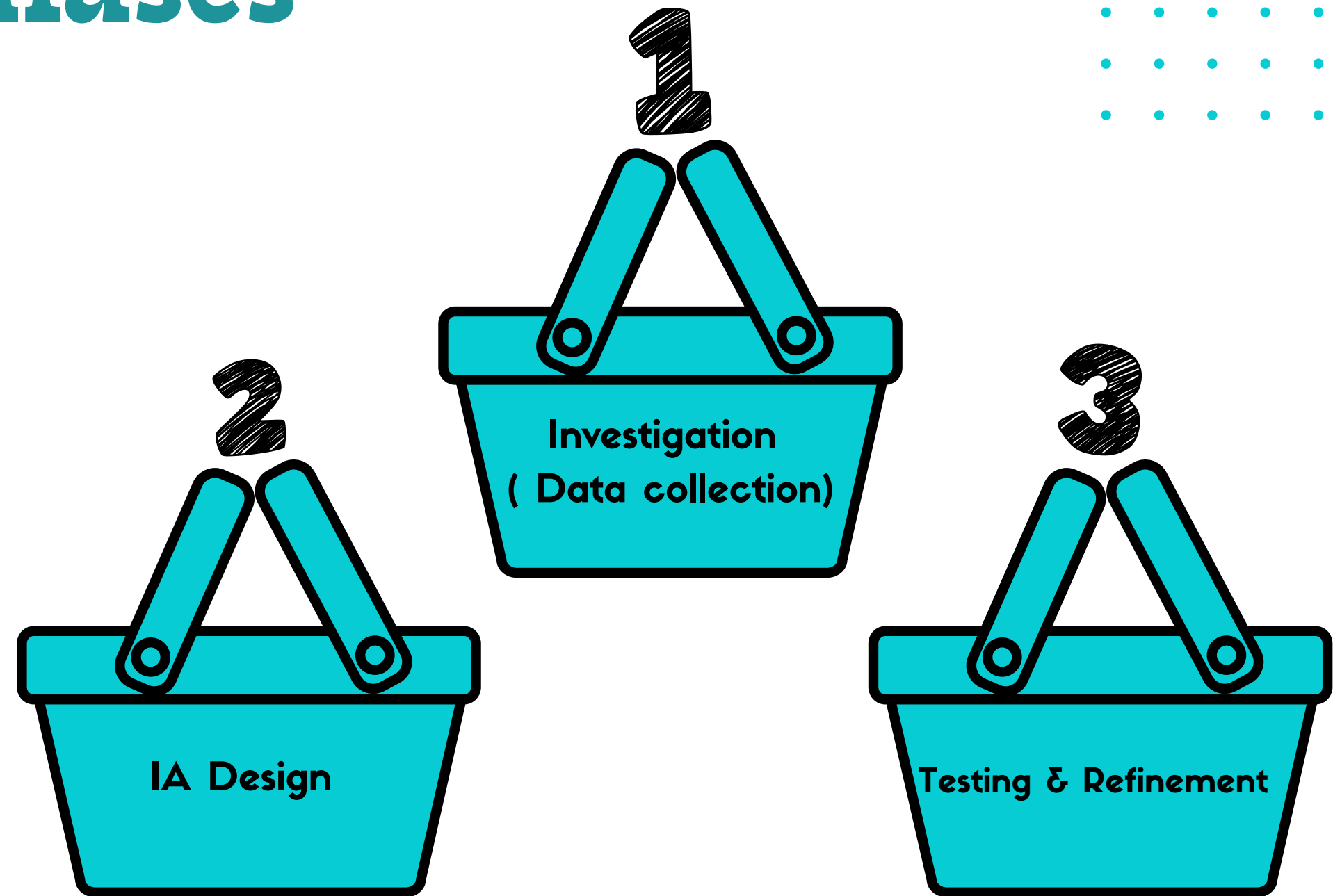


Introduction

This project analyzes the information architecture (IA) of the Ninja app, which is designed to make grocery delivery fast and easy. As people rely more on mobile apps for daily tasks having clear and organized information is very important. The focus is on how products are shown grouped searched for, and how the checkout process works. The goal is to find areas for improvement and suggest ways to make the user experience smoother and more efficient.



Project Phases



Phase 1



Phase 1

This competitive analysis compares two grocery delivery apps in Saudi Arabia: **Nana** and **Ninja**.

The goal is to evaluate each app's user experience, interface structure, and features to understand strengths and weaknesses that can influence our own design.

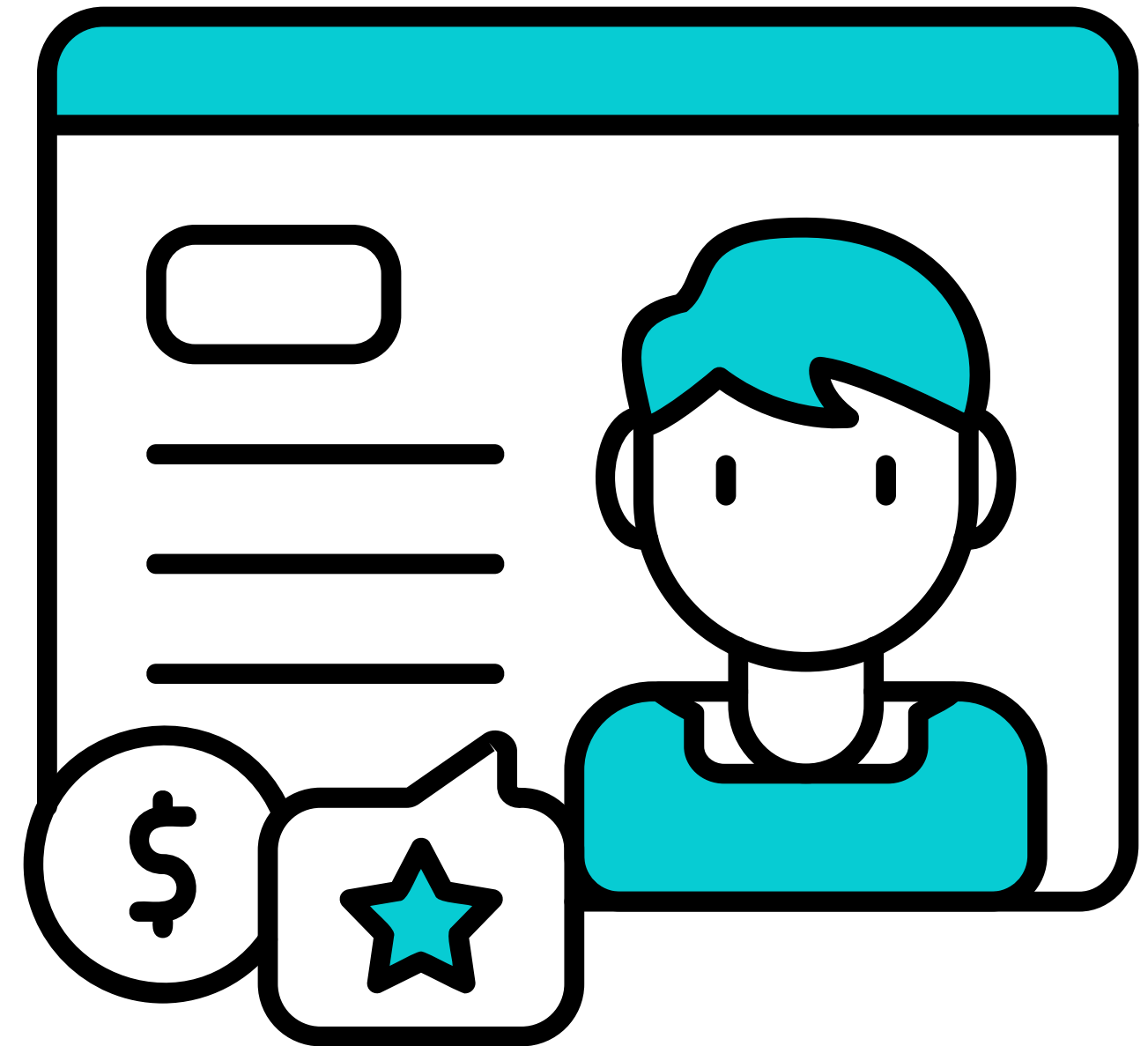


Elements	Ninja App	Nana App
Homepage accessibility	Very good, with clearly visible main section	Acceptable, but lacks clear structure
Search bar usability	Fast, shows suggestions while typing	Sometimes slow, limited results
Clarity of categories	Well-organized (Fruits, Meat, Personal Care)	Categories exist but some are unclear or missing
Search results (accuracy and speed)	Accurate and quick results	Some unrelated results appear
Product display (images, info)	Clear images, full info (price, weight)	image availability but sometimes low quality

Elements	Nana App	Ninja App
Checkout process clarity	clear steps with summary and promo codes	Simple but lacks clear breakdown at final step
Language and localization	Fully localised in Arabic and English	Also supports Arabic and English but some labels may be inconsistent
Navigation between pages	Smooth navigation via bottom ta bar	Uses top tabs; transitions less smooth sometimes
Filters/sorting	Great filtering by brand, price,category	Limited filters; basic sorting available

Personas

It's a fictional character we create based on real data, representing a specific type of our target audience. It helps the team better understand users, design the product and marketing to fit them, and keeps everyone on the same page about who we're building .this for



PERSONA



Grandma Hessa (72 years old)

- Lives alone, has trouble walking far or carrying heavy bags.
- Uses a smartphone but gets confused by complicated apps.
- Buys groceries, medicine, and sometimes cakes or cookies.

Goals:

- ✓ Big, clear buttons – no tiny text!
- ✓ Groceries and medicine delivered to her door.
- ✓ Option to pay with cash (not just cards).
- ✓ Friendly drivers who don't rush her.

Pain points:

- ✗ The app is hard to use (too many steps).
- ✗ Her order gets replaced without asking ("I wanted soft bananas!").
- ✗ Delivery fees feel too expensive.
- ✗ Text is too small and unclear – she struggles to read it.
- ✗ Important items like medicine are hard to find in the menu.

PERSONA



Student Azzam (20 years old)

- Lives in a small apartment with roommates.
- Always busy, forgets to shop, and craves fast food.
- Loves discounts and quick deliveries.

Goals:

- ✓ Cheap, fast meals (pizza, burgers, noodles).
- ✓ Late-night snack delivery (even at 2 AM!).
- ✓ Split bills with friends easily.
- ✓ Live tracking to see where his food is.

Pain points:

- ✗ Delivery takes too long when he's hungry.
- ✗ Extra fees make his cheap meal expensive.
- ✗ No student discounts.
- ✗ Search results aren't always accurate — he wastes time looking.
- ✗ The app homepage feels cluttered and distracting.

PERSONA



Mom Reham(35 years old)

- Works and takes care of two kids.
- Needs groceries, school supplies, and quick dinners.
- Likes planning ahead but sometimes forgets things.

Goals:

- ✓ Big grocery orders (milk, rice, snacks for kids).
- ✓ Option to pick delivery times (like after work).
- ✓ Safe, contactless delivery if she's not home.
- ✓ Notifications if something is out of stock.

Pain points:

- ✗ Delivery comes late when she's cooking dinner.
- ✗ Eggs or bread get damaged in delivery.
- ✗ No option to leave packages with a neighbor.
- ✗ Navigation in the app is sometimes confusing.
- ✗ Some categories and filters are too basic or unclear.

Why competitive Analysis?

We chose the competitive analysis method to better understand the strengths and weaknesses of other delivery apps in the market. This approach allows us to:

- Identify what competitors are doing well and where they are lacking.
- Discover market gaps and opportunities for innovation.
- Benchmark our app's features, user experience, and services against industry leaders.
- Make informed decisions to improve our app and offer a more competitive product.



A decorative graphic consisting of two rows of small blue dots. The top row has 10 dots and the bottom row has 10 dots, arranged in a staggered pattern.

Analysis

Ninja is easy to use with a clear interface and fast, accurate search. Product details are well-presented with high-quality images. However, navigation can be inconsistent, some translations are off, and filters are basic.

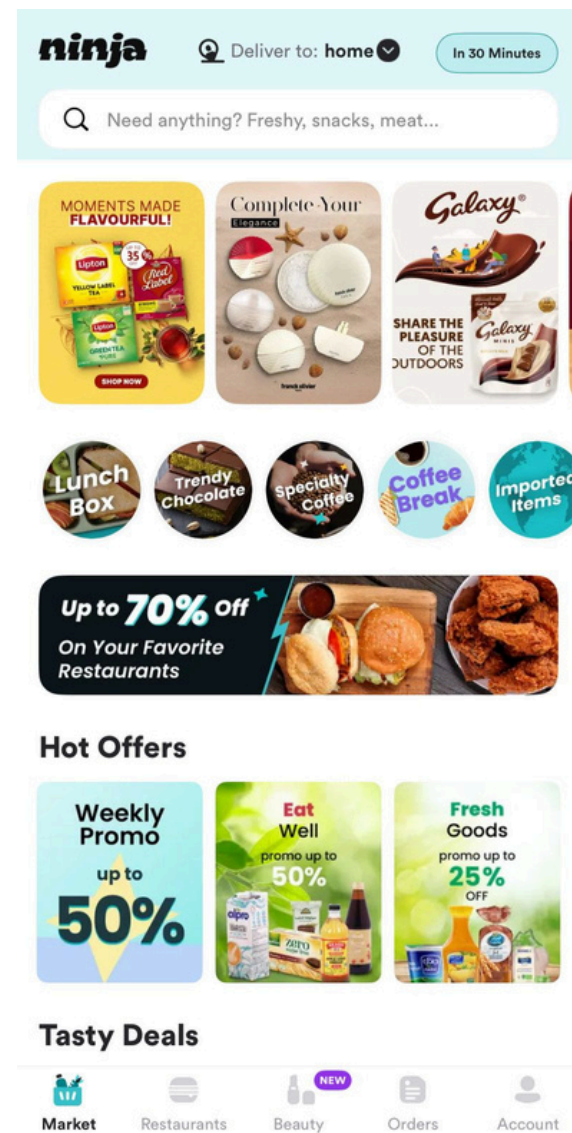
Nana is decent but could improve. The homepage is a bit unorganized, and search results aren't always accurate. Still, it offers a smooth checkout process, supports discount codes, handles both languages well, and provides strong filtering options.

Some examples of Comparison

Homepage

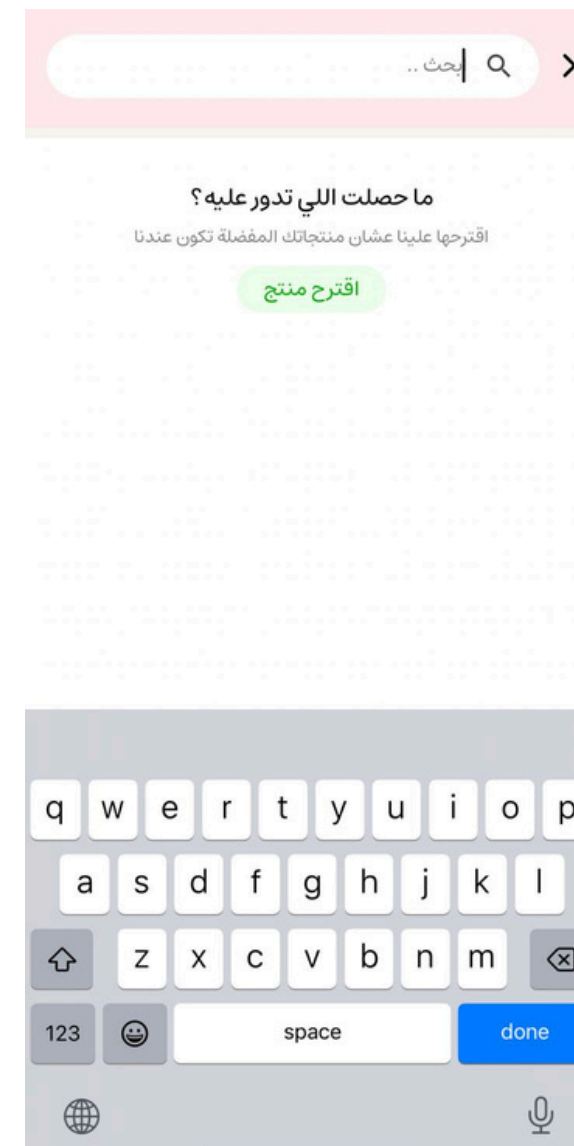


Nana App

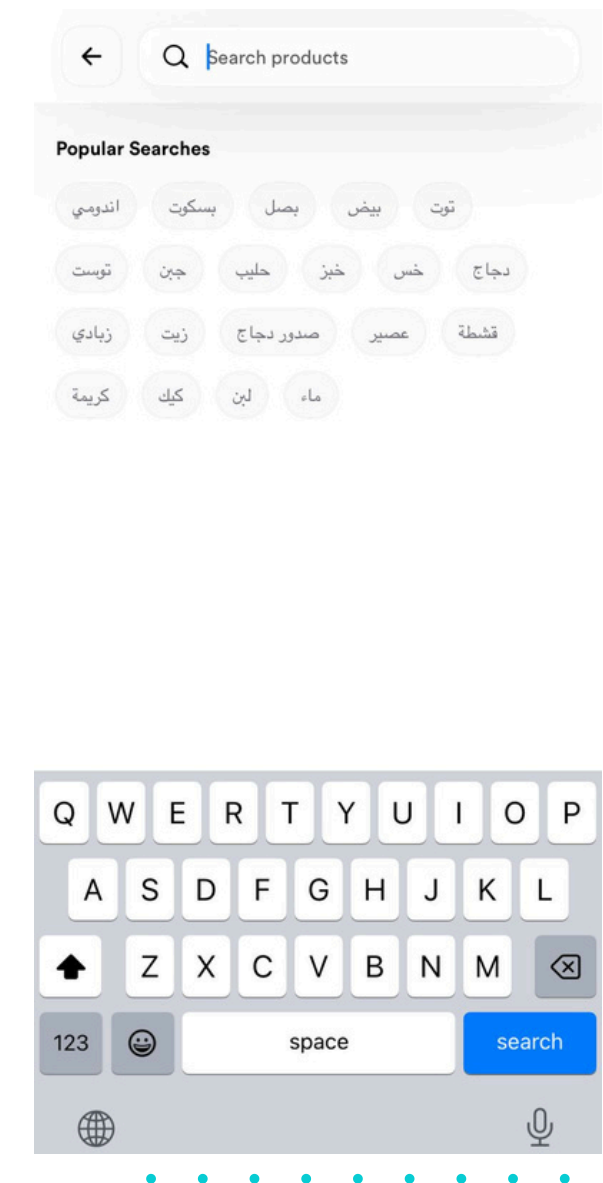


Ninja App

Searchpage



Nana App



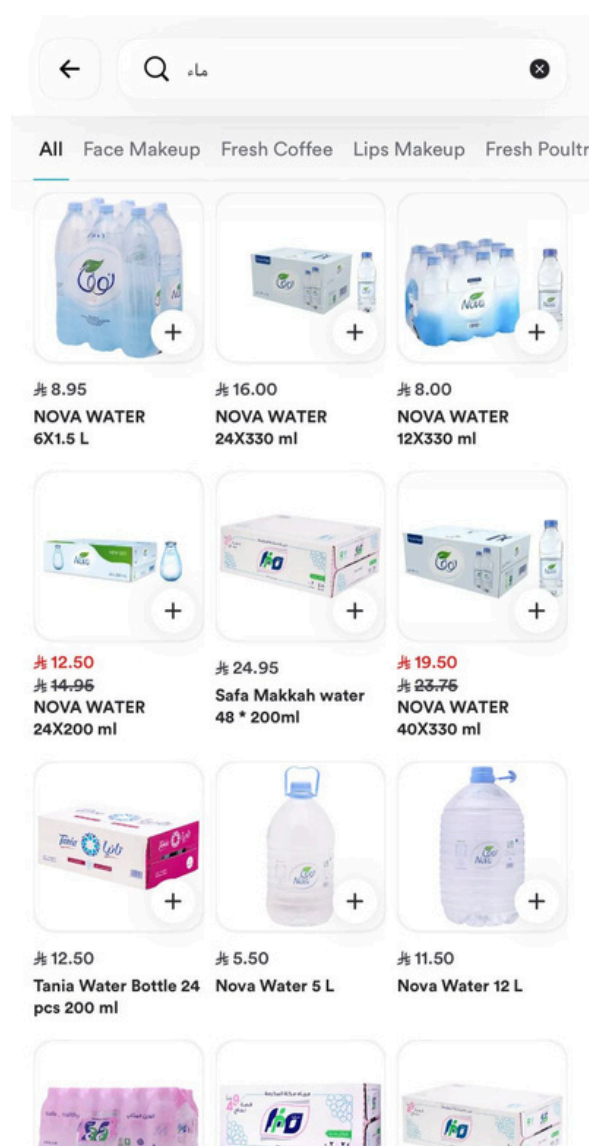
Ninja App

Some examples of Comparison

Product Display page

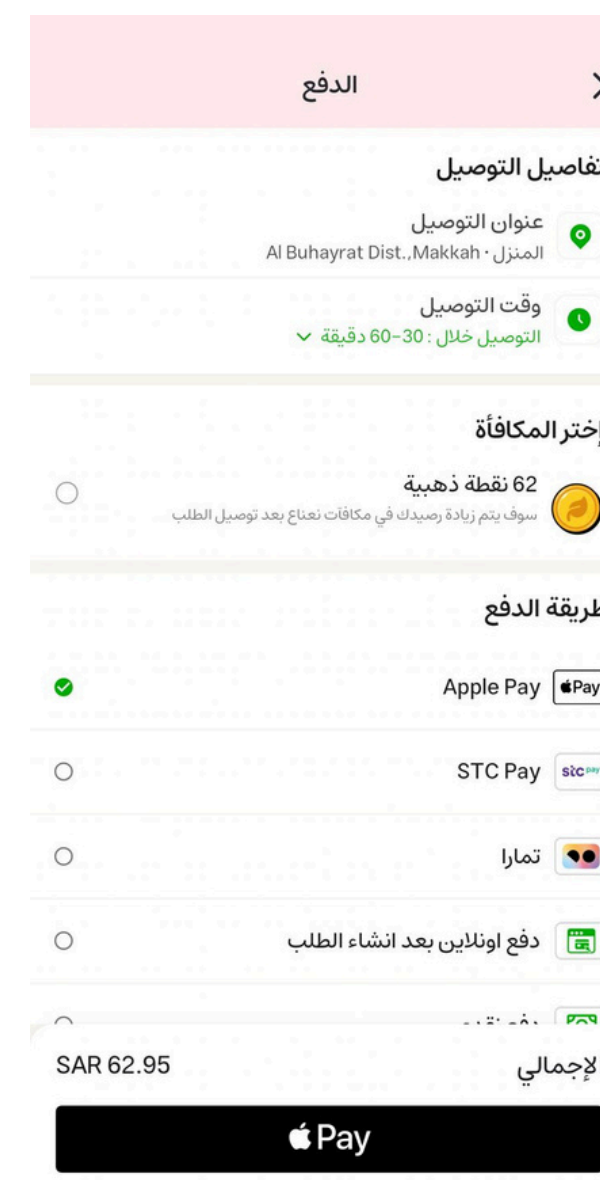


Nana App

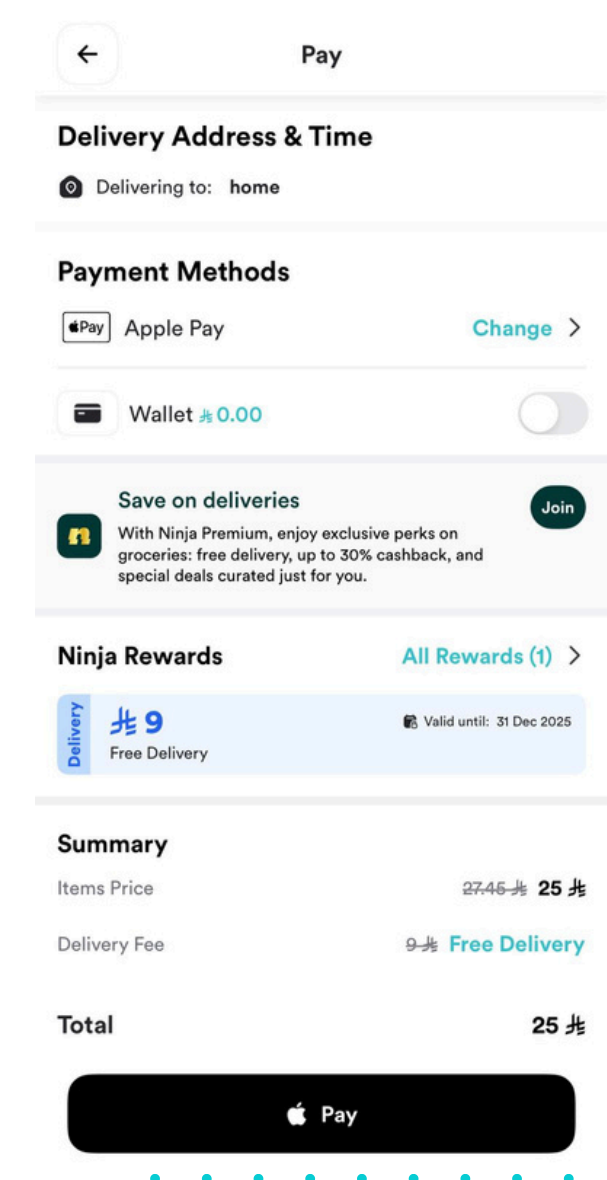


Ninja App

Check out page



Nana App



Ninja App

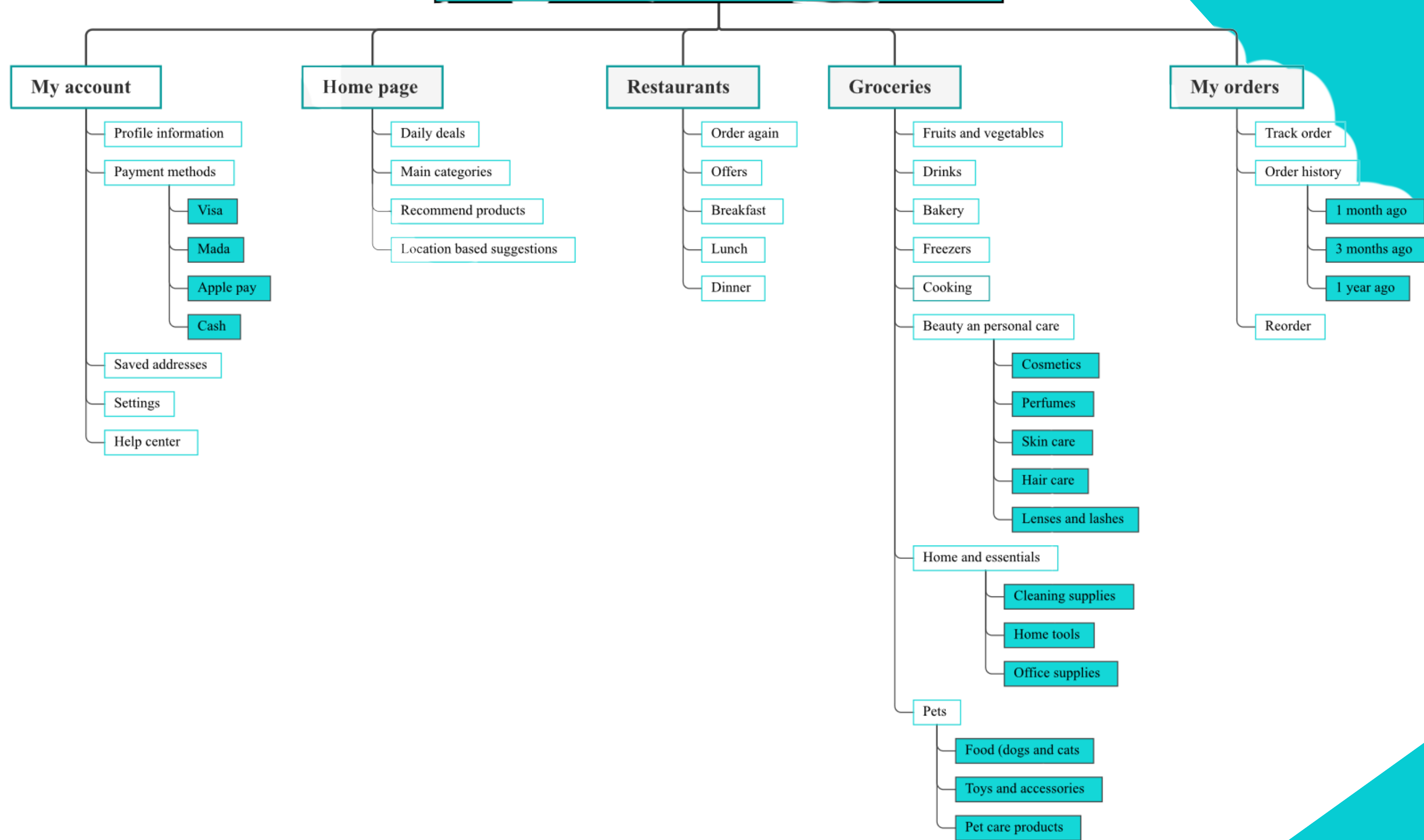
Phase 2



Site map



Site map for ninja application



Color coded



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	Link ID	Page Name	Purpose of the Page	Intended Design Type																				
2	1.0	home page	main landing page	home page template																				
3	1.1	daily deals	daily offer	category listening type																				
4	1.2	main categories	shows the main categories	category listing type																				
5	1.3	recommended products	shows the most recommended products	category listing type																				
6	1.4	Location based suggestions	shows the suggested locations	category listing template																				
7	2.0	groceries	shop groceries	category listing template																				
8	2.1	fruits and vegetables	shop fruits and vegetables	category listing type																				
9	2.2	drinks	shop drinks	category listing type																				
10	2.3	bakery	shop bakeries	category listing type																				
11	2.4	freezers	shop freezers	category listing type																				
12	2.5	cooking	shop cooking	category listing type																				
13	2.6	beauty and personal care	shop makeup and personal care	category listing type																				
14	2.6.1	cosmetics	shop cosmetics	category listing type																				
15	2.6.2	perfumes	shop perfumes	category listing type																				
16	2.6.3	skin care	shop skin care	category listing type																				
17	2.6.4	hair care	shop hair care	category listing type																				
18	2.6.5	lenses and lashes	shop lenses and lashes	category listing type																				
19	2.7	home and essentials	shop home and essentials	category listing type																				
20	2.7.1	cleaning and supplies	shop cleaning tools and supplies	category listing type																				
21	2.7.2	home tools	shop home tools	category listing type																				
22	2.7.3	office supplies	shop offices supplies	category listing type																				
23	2.8	pets	shop for pets	category listing type																				
24	2.8.1	food (dogs and cats)	shop food for dogs and cats	category listing type																				
25	2.8.2	toys and accessories	shop toys and accessories for dogs and cats	category listing type																				
26	2.8.3	pet care products	shop care products for dogs and cats	category listing type																				
27	3.0	restaurant	main restaurant page	main page template																				
28	3.1	order again	shows the previous orders to order it again	category listing type																				
29	3.2	offers	shows the offers	category listing type																				
30	3.3	breakfast	shows the breakfast restaurants	category listing type																				
31	3.4	lunch	shows the lunch restaurants	category listing type																				
32	3.5	dinner	shows the dinner restaurants	category listing type																				
33	4.0	my orders	shows the previous orders	main page template																				
34	4.1	track orders	track orders	category listing type																				
35	4.2	track order history	shows previous orders	category listing type																				
36	4.2.1	1 months ago	shows orders a month ago	category listing type																				
37	4.2.2	3 months ago	shows orders a 3 months ago	category listing type																				
38	4.2.3	1 year ago	shows orders a 1 year ago	category listing type																				
39	4.2.4	reorder	reorder previous orders	category listing type																				
40	5.0	my account	Special for account information	main page template																				
41	5.1	profile information	Shows profile information	category listing type																				
42	5.2	payment methods	Shows payment methods	category listing type																				
43	5.2.1	visa	Shows payment methods	category listing type																				
44	5.2.2	mada	Shows payment methods	category listing type																				
45	5.2.3	apple pay	Shows payment methods	category listing type																				
46	5.2.4	cash	Shows payment methods	category listing type																				
47	5.3	saved address	Shows the saved address	category listing type																				
48	5.4	settings	Shows the settings	category listing type																				
49	5.5	help center	Shows the help center	category listing type																				

Card sorting



Closed Card Sorting Report – Ninja App

Overview:

A Closed Card Sorting study was conducted using the Optimal Workshop platform to evaluate the information architecture of the “Ninja” shopping app. In this type of study, participants were provided with predefined categories and asked to sort individual items (cards) into the most appropriate category.

Study Details:

- Platform: Optimal Workshop (OptimalSort tool)
- Method: Closed Card Sorting
- Number of cards: Approximately 35
- Number of categories: 5 (My Account, Home Page, Restaurants, Groceries, My Orders)
- Study language: English (US)
- Participant location: 100% from Saudi Arabia

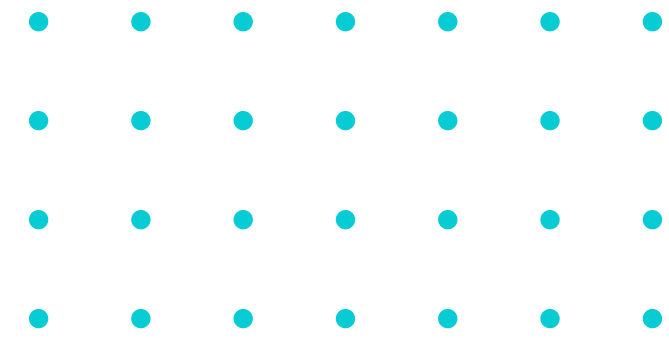
Participants Summary:

- Invited participants: 7
- Completed the study: 5 participants
- Completion rate: Approximately 71%
- Card sorting completion: 100% of the cards were sorted by each participant
- Median time to complete: 6 minutes and 31 seconds
 - Shortest time: 5:35
 - Longest time: 7:27
- Question responses: Participants submitted responses to follow-up questions, providing qualitative feedback on the sorting process and category clarity.

Initial Observations:

- All five participants completed the sort successfully.
- Categories such as “My Orders” and “Restaurants” were consistently understood.
- Some overlap or confusion was noted in categories like “Groceries” and “Beauty & Personal Care.”
- Participant feedback through the open-ended questions revealed insights on potential improvements for labeling and grouping.

Analysis & Recommendations:



Based on the results, the following structure is recommended for the app:

- Main Categories:
 - Breakfast
 - Lunch/Dinner
 - Snacks / Coffee
 - Healthy Food
- Additional Filters:
 - Restaurant type (e.g., Burger, Pizza, Juices, etc.)
 - Delivery speed
 - User ratings

Final Categorization Based on Card Sorting Results

Main Categories:

Participants consistently grouped food items into key daily meal times. As a result, the following main categories are recommended:

- **Breakfast:** includes items typically consumed in the morning such as eggs, fowl, or croissants.
- **Lunch / Dinner:** includes heavier meals such as burgers, pizza, shawarma, or rice dishes.
- **Snacks / Coffee:** consists of light bites and beverages including juices, snacks, and various types of coffee.
- **Healthy Food:** includes health-conscious options such as salads, smoothies, or fruit-based meals.

Suggested Filters:

To allow users more control in their browsing and searching experience, the following filters were frequently mentioned or implied during the card sorting session:

- **Restaurant Type** (e.g., Burger, Pizza, Juices, etc.)
- **Delivery Speed** (e.g., Fast, Moderate, Slow)
- **User Ratings** (based on customer reviews and star rankings)

Navigation

Search



Navigation & Search

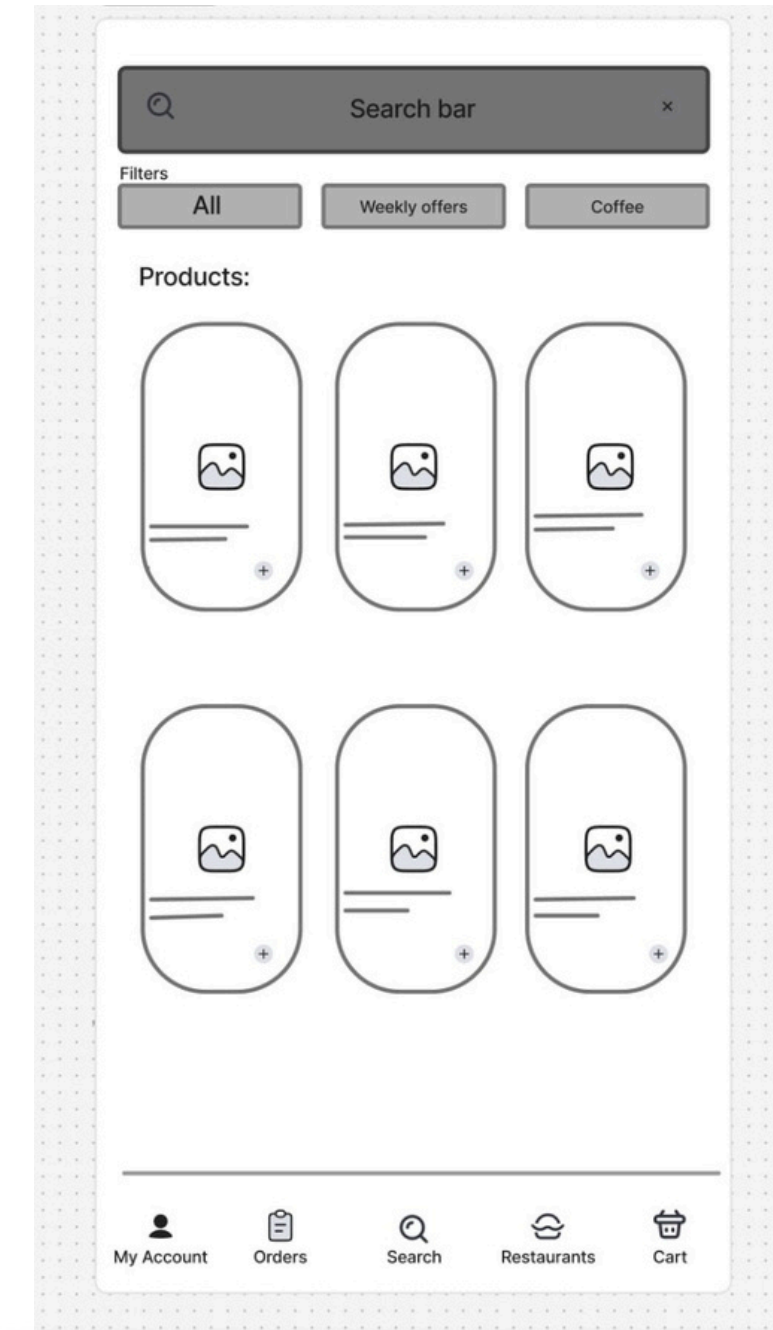
Description:

The app provides a smart search bar that supports autocomplete and synonyms, making it easier for users to find product.

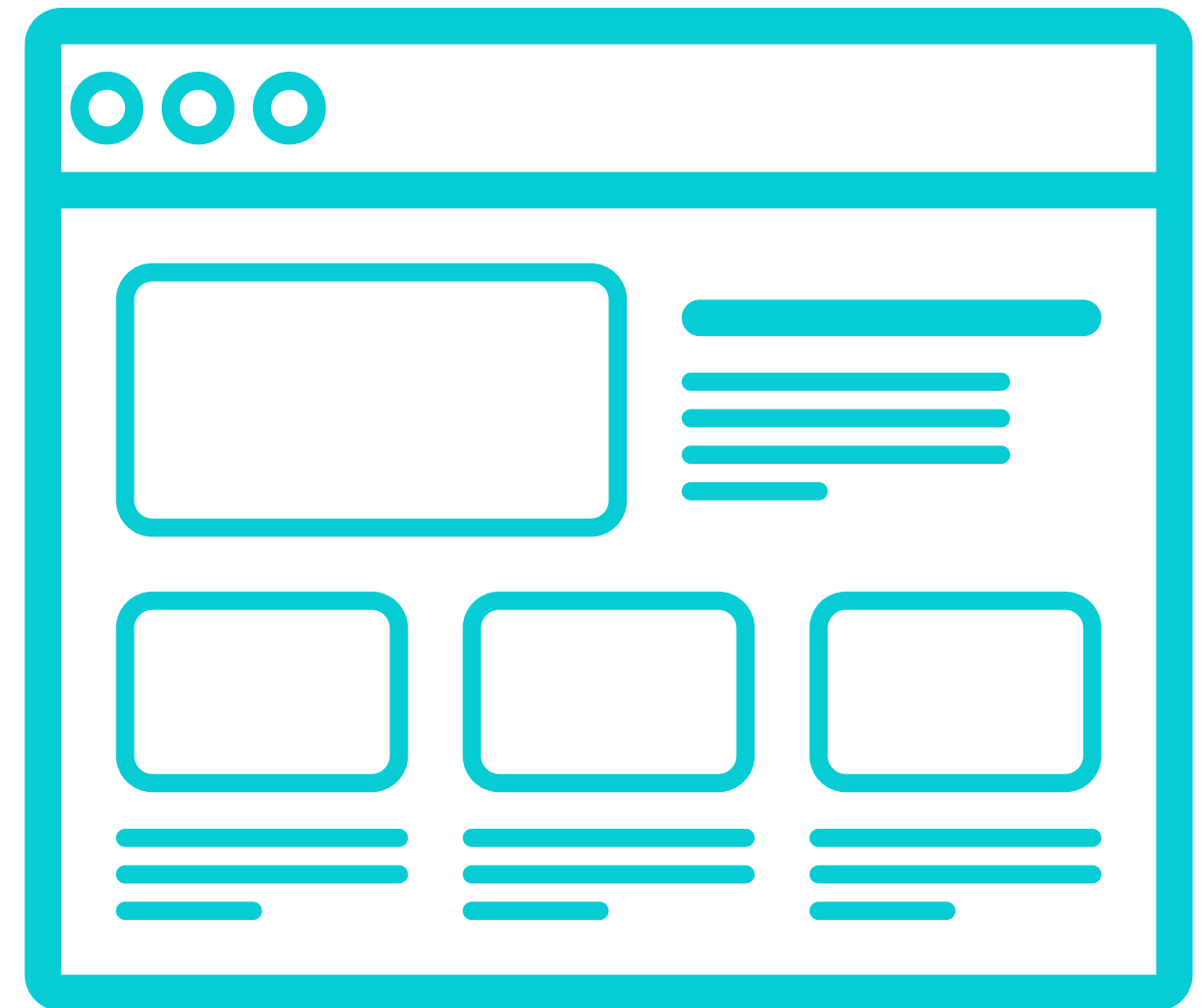
Below the search bar, there are horizontal filters such as: All, Weekly offer, Coffee.... allowing users to narrow down search results based on their preference.

Products are displayed as cards showing an image, product name, price, and an add(+) button. Some cardamom include special offers (e.g., Buy 5 for a special price).

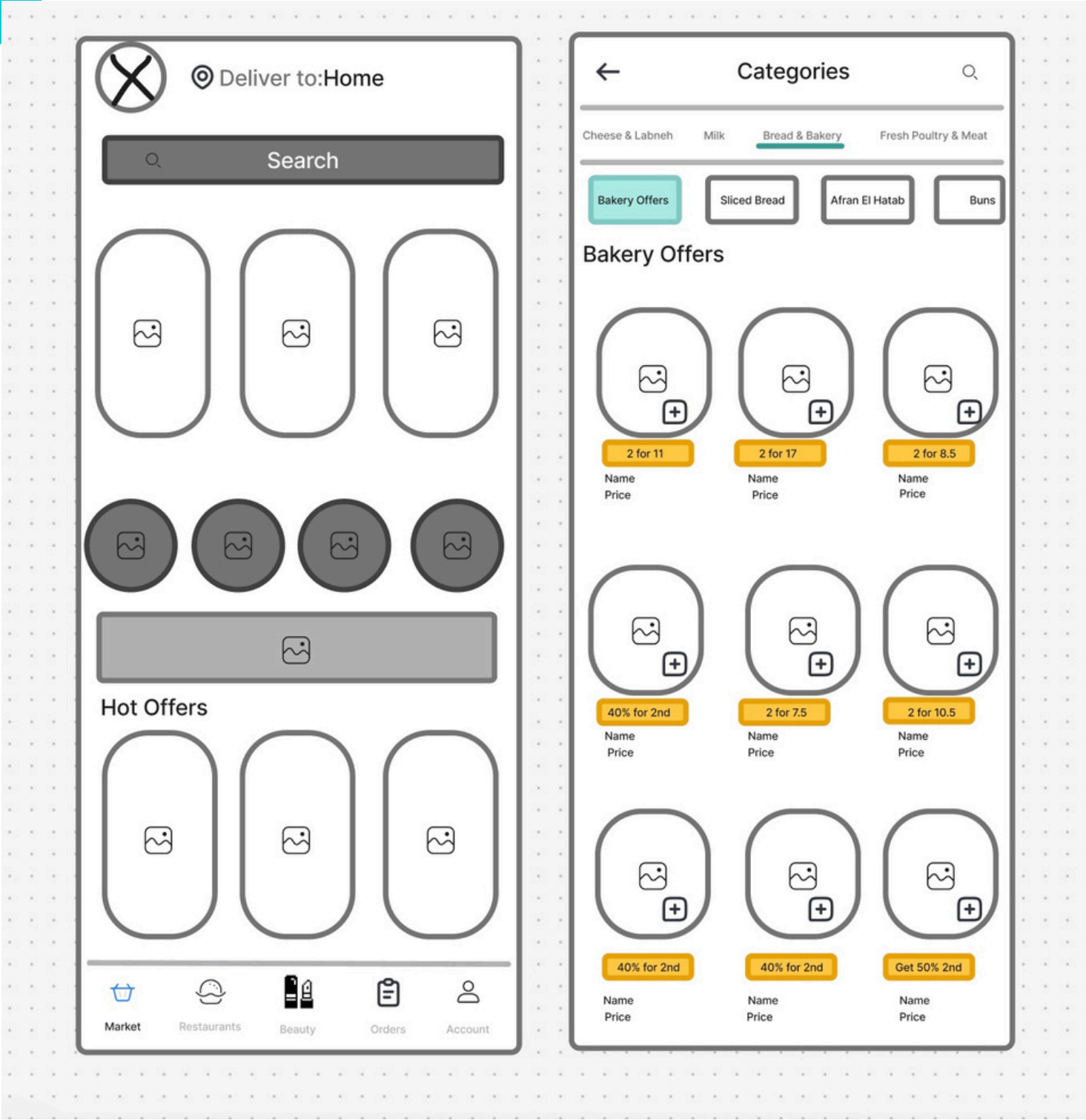
At the bottom, a navigation bar include: My Account, Orders, Search, Restaurants, and Cart.



Wireframing (low-Fi)



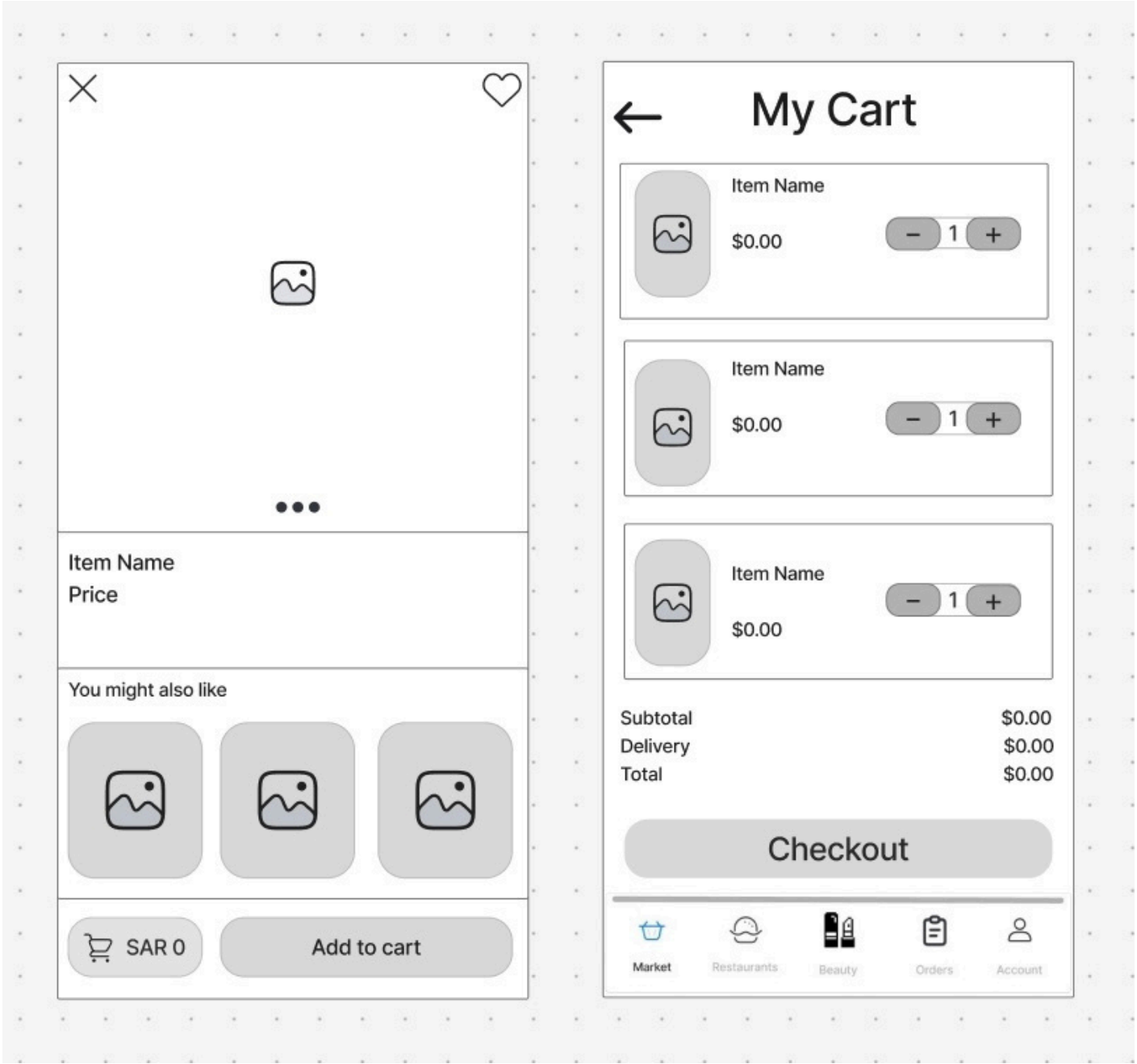
Home Page & Product List Wireframe



Home page

Grocery page

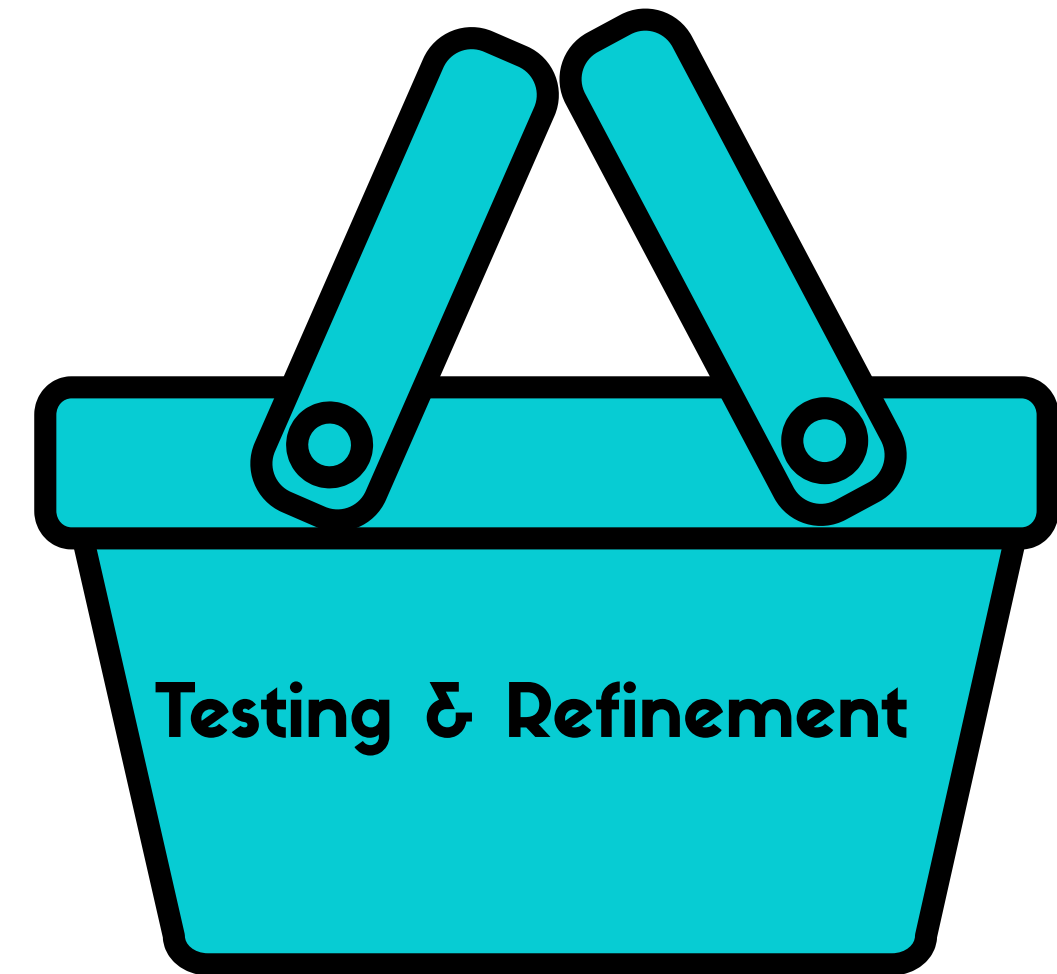
Cart Pay Wireframe

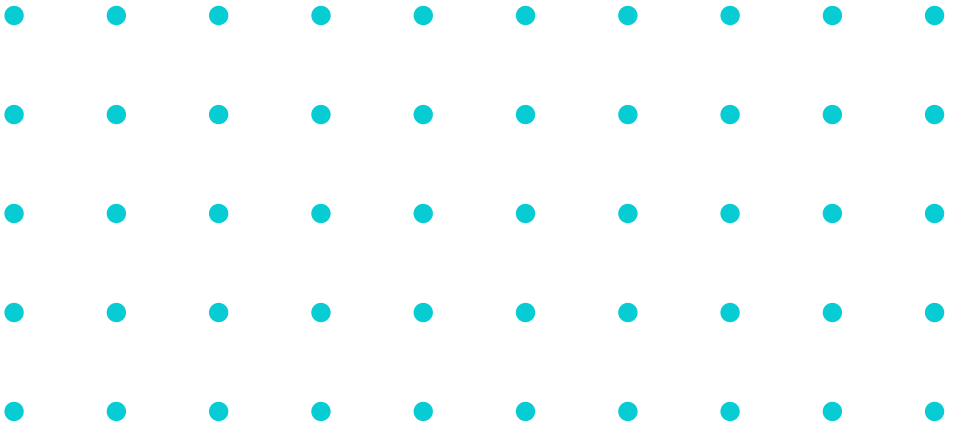


product select page

cart page

Phase 3





Tree testing



Tree testing

In this stage, I conducted a Tree Testing activity using the [UXtweak](#) platform to evaluate the clarity of the structure of the “Ninja” app and how easily users can navigate it.

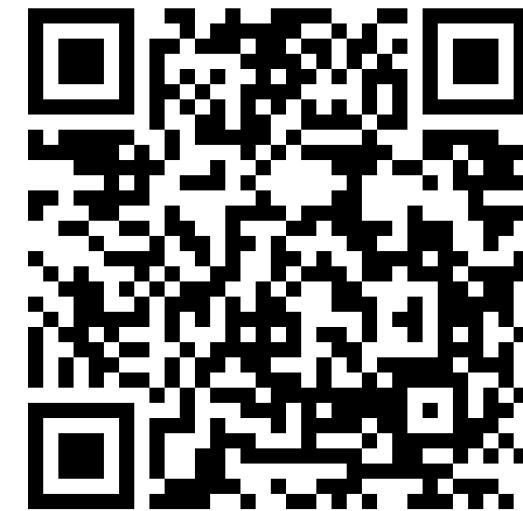
I created two tasks that represent real user scenarios:

[Adding a payment method \(e.g., Mada or Visa\)](#)

[Finding cleaning supplies for the home](#)

The site map of the app was uploaded to the platform, and each task was linked to its correct path within the tree. The test was then shared with users.

User selections and navigation behavior were recorded and will be used later to analyze the effectiveness of the app’s structure and identify areas for improvement.



Usability Testing Observations

Number of participants: 7

User 1: Quick grocery order

Goal: Test how easy it is to place a grocery order

Post-Test Feedback:

- Overall: Frustrating at first
- Easy: Once groceries section found
- Hard: Labels unclear (Add/Chat)
- Suggestions: Use clearer icons/labels

What You Observe	Details	Time
Did the user know where to start?	No-User scrolled up and down, said Where are groceries?	1 minute 25 seconds
Were their steps clear and logical?	Not really - Clicked "Chat" first	
Did they hesitate or stop?	Yes-Paused for 10 seconds	
Did they backtrack?	Yes - 2 times	
Wrong clicks?	Yes-Clicked "Add" thinking it's the grocery section	
Quotes	This is confusing where the groceries section?	

Usability Testing Observations

User 2: Reordering a Past Purchase

Goal: Can the user reorder something
they bought before

Post-Test Feedback:

- Overall: Confusing
- Easy: After finding the right button
- Hard: Naming of “Reorder” and “Track Order”
- Suggestions: Separate buttons or clearer explanation

What You Observe	Details	Time
Did the user know where to start?	Kind of - User looked in Track Order	2 minutes 10 seconds
Steps clear?	Not at first - Confused between reorder and track order	
Hesitation?	Yes - Opened wrong section, then asked for help	
Did they backtrack?	Yes - Went back and forth between pages	
Wrong clicks?	Yes-Opened “Track Order” first	
Quotes	Aren’t they the same thing?	

Usability Testing Observations

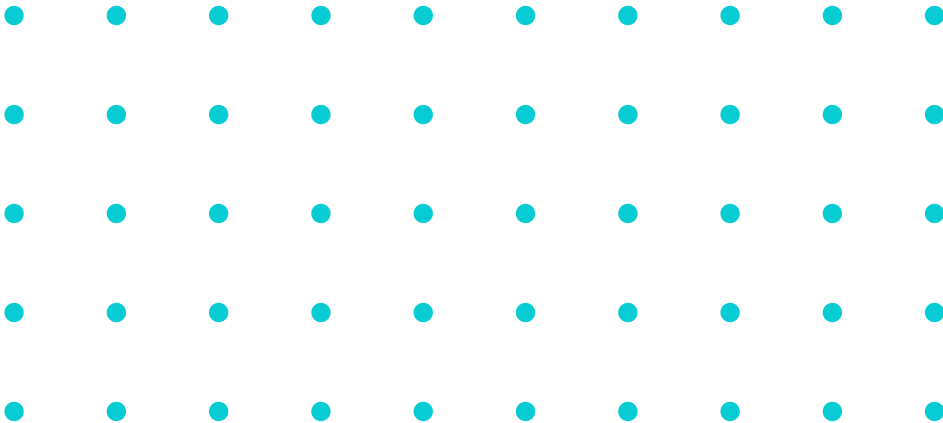
User 3: Editing Profile & Payment Info

Goal: Can the user change their saved address and card info

Post-Test Feedback:

- Overall: Frustrating and slow
- Easy: Once in the right menu
- Hard: Structure of settings unclear
- Suggestions: Add shortcut to addresses and payment under "Profile"

What You Observe	Details	Time
Did the user know where to start?	No-User searched settings blindly	2 minutes 40 seconds
Steps clear?	No-User clicked My Account then Orders by mistake	
Hesitation?	Yes - Looked around for 30+ seconds	
Did they backtrack?	Yes - Opened 3 wrong menus	
Wrong clicks?	Yes-Didn't find "Saved Addresses" easily	
Quotes	I thought this would be under profile	



Usability & Tree Testing Analysis



Usability Testing Summary

The usability test was structured around 3 real-life scenarios:

Scenario 1: Quick Grocery Order

- Goal: Test how easy it is to place a grocery order.
- Insight: Some users struggled to find the “Groceries” section quickly. The labels like “Add” and “Checkout” weren’t always clear.
-

Scenario 2: Reordering a Past Purchase

- Goal: Check if users can easily reorder from past purchases.
- Insight: Many users confused “Reorder” with “Track order”. It wasn’t obvious which option does what.
-

Scenario 3: Editing Profile & Payment Info

- Goal: Test if users can update their address and payment details easily.
- Insight: Several users had trouble finding “Saved Addresses” or navigating to the right settings.

Common Pain Points Across Scenarios

1. **Unclear Terminology** – Words like Reorder and Track Order were often misunderstood.
2. **Navigation Issues** – Users often got lost or took long paths to reach their goal.
3. **Too Many Steps** – Tasks like reordering or editing info required more steps than expected.
4. **Lack of Guidance** – No helpful prompts for first-time users.

Tree Testing Summary

- **Participants:** 10 (6 completed, 4 dropped out)
- **Success Rate:** 58.3% completed tasks correctly.
- **Directness:** 66.7% completed tasks without backtracking.
- **Median Task Time:** ~53 seconds

Task Performance Breakdown

Task	Success	Failure
Task 1	~45%	~55%
Task 2	~30%	~70%

Interpretation:

- Many users didn't reach the correct answers directly.
- A high failure rate suggests confusing navigation structure or unexpected menu labels.

Recommendations

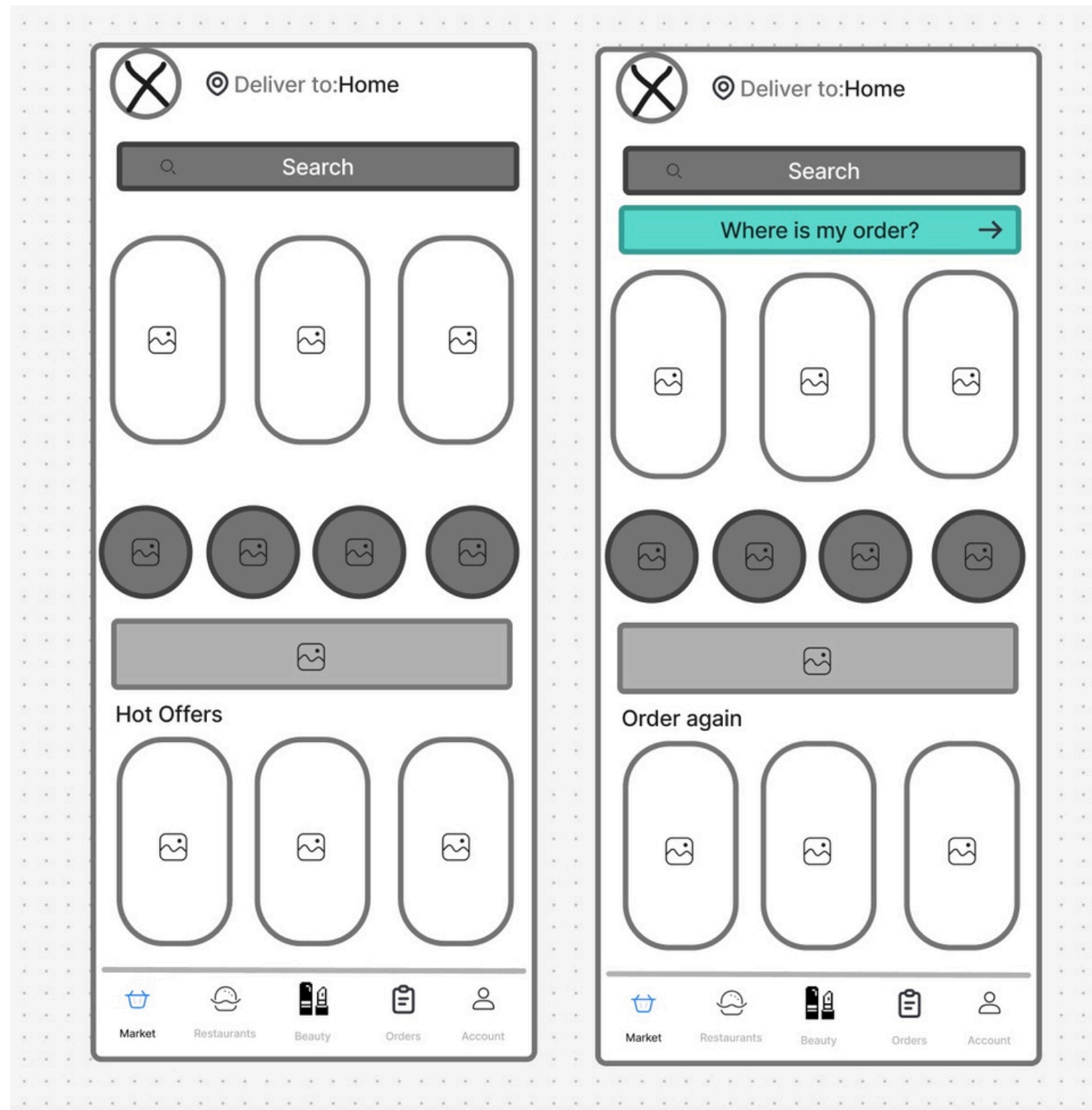
- Simplify and clarify labels (e.g., change “Track Order” to “Order Status”).
- Streamline user flows to reduce the number of steps per task.
- Improve tree structure to align more closely with user expectations.
- Add helper text or tooltips for first-time users.

Here's a visual comparison of the success and failure rates across the two Tree Testing tasks :



Final interface

Unclear Terminology – Words like Reorder and Track Order were often misunderstood.



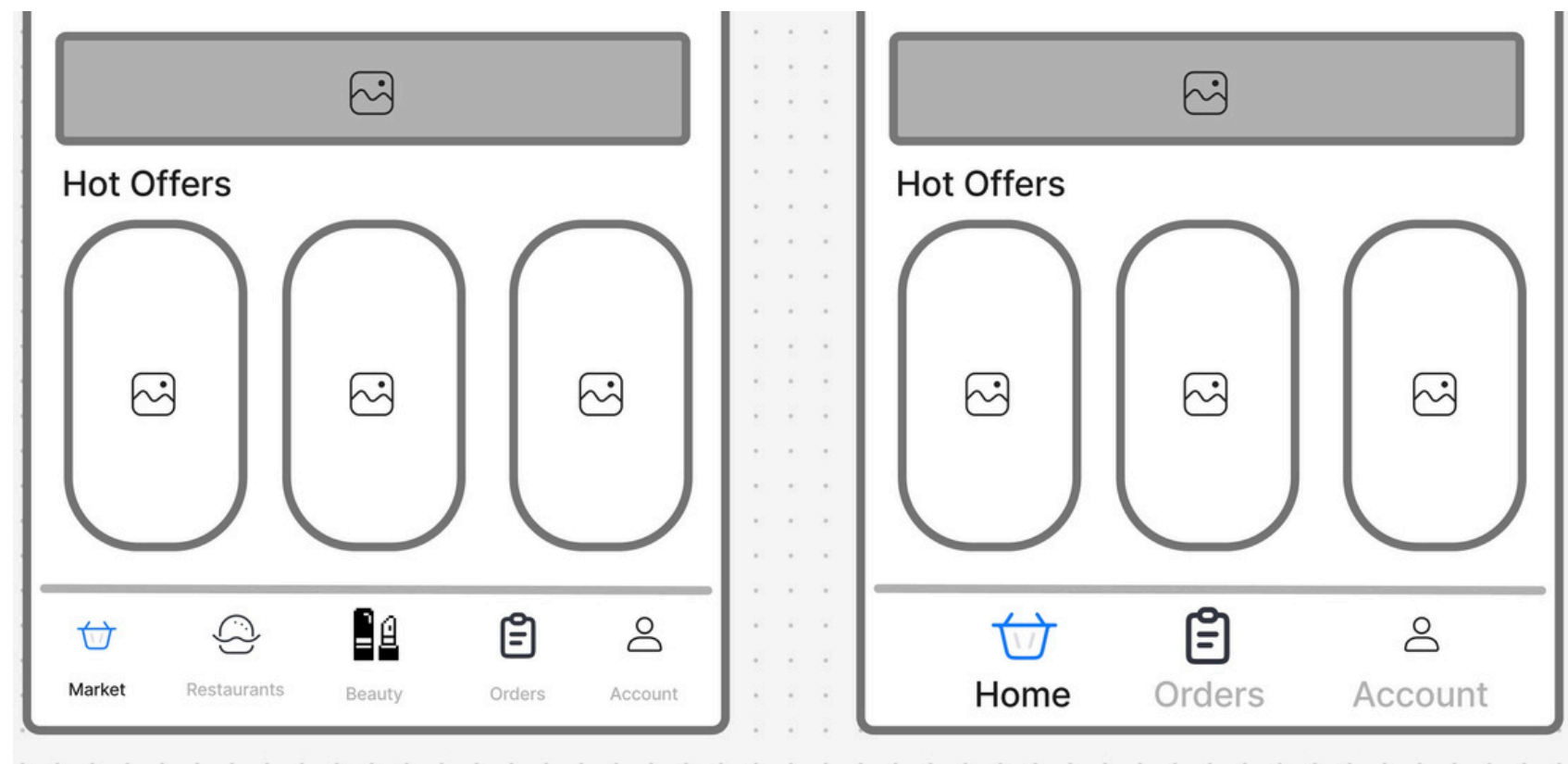
Before

After

We added “where is my order?”
and “order again” instead of “Hot offers”

Final interface

Navigation Issues – Users often got lost or took long paths to reach their goal..



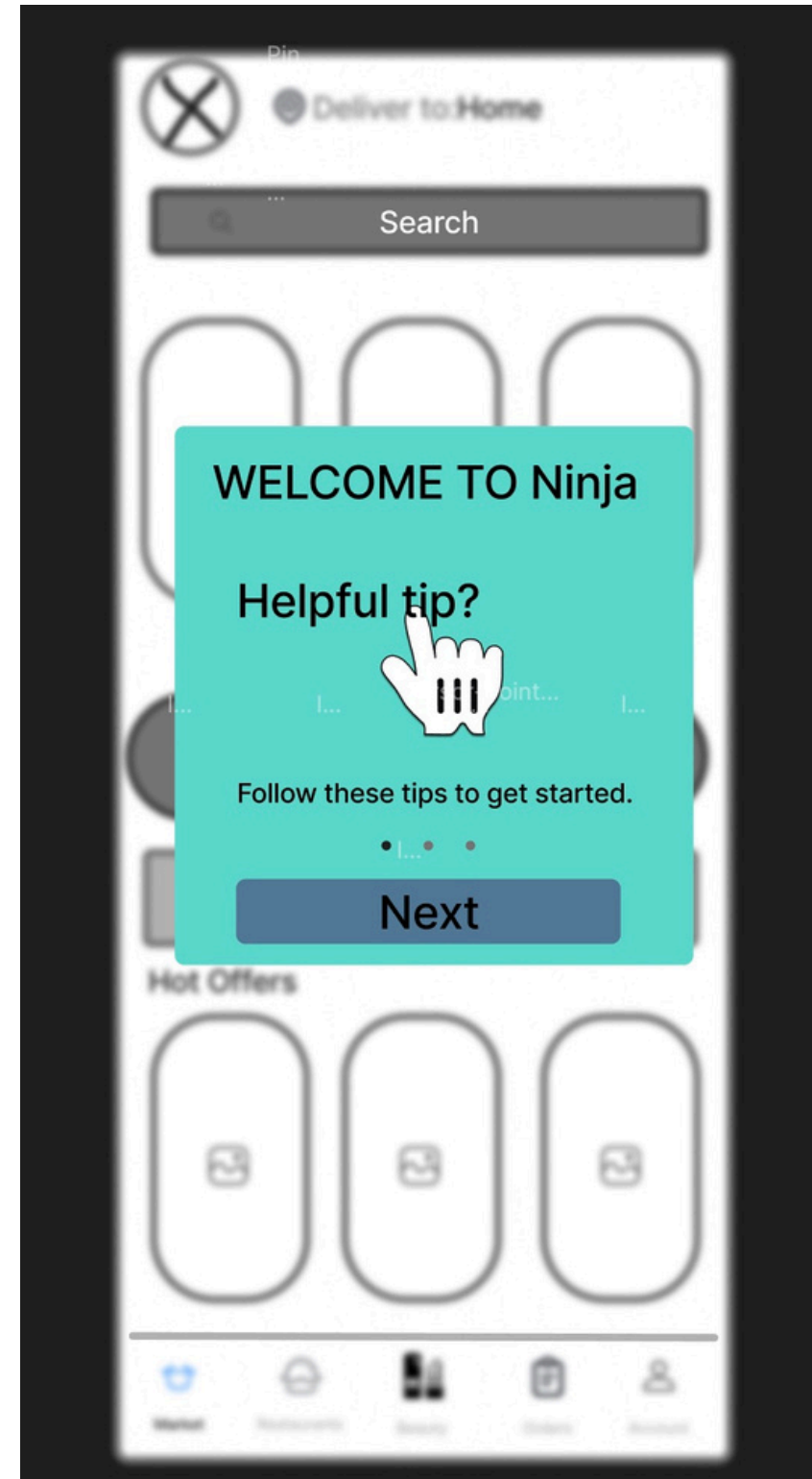
the solution is to limits the navigation bar and add only the important sections

Before

After

Final interface

Lack of Guidance – No helpful prompts for first-time users.



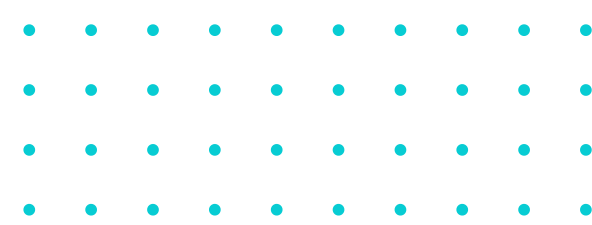
Added “Helpful structure” at the beginning of the app

Conclusion

This project gave a full review of how the Ninja grocery app is organized. We looked at important parts like how products are grouped, how the search works, and how users pay for their orders. We found things the app does well and other things that can be better. A good app structure helps users have an easy and happy experience, especially in fast services like grocery delivery. Our ideas and suggestions can help make the app easier to use, better to move around in, and more enjoyable. This project shows how important it is to design apps that focus on what users need to make digital services successful.



Table Task



	A	B	C	D
1	Phase	Task	Assigned To	Status
2	Phase 1	IA Comparison	Hind, Dhuha	Completed
3	Phase 1	Personas	Hajar, Sham	Completed
4	Phase 1	Analysis	Azoof, Marah	Completed
5	Phase 2	Site Map (Excel Sheet)	Hajar	Completed
6	Phase 2	Card Sorting	Sham, Azoof	Completed
7	Phase 2	Navigation Search	Dhuha	Completed
8	Phase 2	Wireframing	Hind, Marah	Completed
9	Phase 3	Tree Testing	Dhuha	Completed
10	Phase 3	Usability Testing	Hajar, Hind	Completed
11	Phase 3	Usability & Tree Testing Analysis	Sham, Azoof	Completed
12	Phase 3	Iterations	Marah	Completed
13	Phase 4	Final Phase (All members involved)	All Members	Completed

